

Multiple Choice

1. **Answer:** C

Options: A) 3 atm; B) 9 atm; C) 5 atm; D) 6 atm; E) 10 atm

Note: According to Boyle's Law, which states that pressure and volume are inversely proportional at constant temperature, compressing 5 liters of gas at 1 atm to 1 liter would require a pressure of 5 atm.

2. **Answer:** E

Options: A) $7.50 \times 10^{-3}\%$; B) $2.50 \times 10^{-3}\%$; C) $0.75 \times 10^{-3}\%$; D) $2.00 \times 10^{-3}\%$; E) $1.25 \times 10^{-3}\%$

Note: The correct answer is derived from the Taylor series expansion of e^x around $x = 0$, which shows that e^x is approximately equal to $1 + x + \frac{x^2}{2}$ for small values of x , and thus the percentage difference can be calculated as $\frac{e^x - (1+x)}{1+x} \times 100\%$, yielding a result of approximately $1.25 \times 10^{-3}\%$ for $x = 0.0050$.

3. **Answer:** A

Options: A) Add 67.7 mL of 12.0 M acid to the flask, and then dilute it with 33.3 mL of water.; B) Add 33.3 mL of water to the flask, and then add 66.7 mL of 12.0 M acid.; C) Add 40 mL of water to the flask, and then add 60 mL of 12.0 M acid.; D) Add 60 mL of 12.0 M acid to the flask, and then dilute it with 40 mL of water.; E) Add 50 mL of 12.0 M acid to the flask, and then dilute it with 50 mL of water.

Note: The correct answer is based on the dilution equation, which states that the product of the concentration and volume of the stock solution must equal the product of the concentration and volume of the diluted solution, allowing for the calculation of the required volumes to achieve a final concentration of 4.0 M from a stock solution of 12.0 M.

4. **Answer:** D

Options: A) pH is 10, pOH is 4; B) pH is 4, pOH is 10; C) pH is 3, pOH is 3; D) pH is 4, pOH is 4; E) pH is 1, pOH is 1

Note: The pH of 0.0001 N HCl is calculated to be 4 due to the complete ionization of the strong acid, resulting in a hydrogen ion concentration of 0.0001 M, while the pOH of 0.0001 N KOH is also 4 because the concentration of hydroxide ions is equal, leading to a neutral solution at 25°C where pH + pOH equals 14.

5. **Answer:** A

Options: A) - 13.61 eV; B) $1.6022 \times 10^{-19}C$; C) $-2.1802 \times 10^{-18}J$; D) $9.1095 \times 10^{-31}kg$

Note: The correct answer of -13.61 eV for the energy of a hydrogen atom in its ground state is derived from

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because, according to the law of conservation of mass, the total mass of the reactants in a chemical reaction must equal the total mass of the products, meaning that the mass of V_2O_5 produced cannot exceed the combined mass of VO and Fe_2O_3 initially present.

7. **Answer:** True

Options: A) True; B) False

Note: The atomic weight of Calcium is approximately 40.2 g/mol because it is calculated based on the weighted average of the isotopes of Calcium found in nature, which is consistent with its position in the periodic table and its atomic mass unit.

8. **Answer:** False

Options: A) True; B) False

Note: An ester is formed by the reaction of an alcohol with an acid, not an amine, which instead reacts with acids to form amides.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the quadratic equation $x^2 - 7x + 10$ can be factored as $(x - 3)(x - 4) = 0$, yielding the solutions $x = 3$ and $x = 4$.

10. **Answer:** False

Options: A) True; B) False

Note: London dispersion forces arise from temporary dipoles created by the fluctuations in electron distribution within molecules rather than from permanent dipoles associated with molecular shapes.

Long Form Response

11. **Answer:** To calculate the empirical formula of a compound composed of 51.9% carbon, 4.86% hydrogen, and 43.2% bromine, we first convert the percentages to grams, assuming a 100 g sample, which gives us 51.9 g of carbon, 4.86 g of hydrogen, and 43.2 g of bromine. Next, we determine the number of moles of each element by dividing by their respective atomic masses: carbon (12.01 g/mol), hydrogen (1.008 g/mol), and bromine (79.90 g/mol). This yields approximately 4.32 moles of carbon, 4.81 moles of hydrogen, and 0.54 moles of bromine. We then divide each mole value by the smallest number of moles (0.54) to simplify the ratio, resulting in an empirical formula of C_6H_7Br . The significance of the empirical formula lies in its ability to provide fundamental insight into the composition of the compound, essential for understanding its chemical behavior and reactivity in various contexts.

Note: correct because it accurately applies the method of converting percentage composition to grams, calculates the number of moles for each element using their atomic

masses, and ultimately allows for the determination of the simplest whole-number ratio of the elements, which defines the empirical formula of the compound.

12. **Answer:** An average adult loses approximately 2.10 gallons of liquid water annually through various physiological processes, including urine, sweat, fecal matter, and expired air. Over a span of 60 years, this totals around 126 gallons of water lost. This significant volume of water loss underscores the importance of maintaining adequate hydration to support metabolic functions and overall health. The implications of these losses highlight the necessity for individuals to adjust their fluid intake according to activity level, environmental conditions, and personal health status to prevent dehydration and ensure optimal physiological performance. Understanding these hydration needs is critical for optimizing health outcomes and performance in both everyday life and specialized settings.

Note: correct because it accurately calculates the total volume of water lost by an average adult over 60 years, emphasizing the critical need for adequate hydration to counterbalance these losses and support essential bodily functions.